



Lecture 3

Introduction to CNC Milling programming (part 2)

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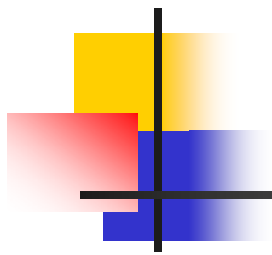
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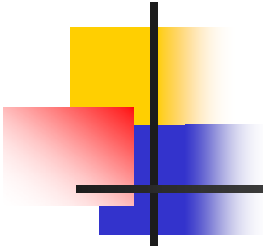
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2- PROGRAMMING PREPARATORY FUNCTIONS



Feed rate/speed programming, G93 to G97

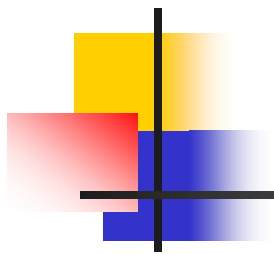
F: Feed rate

- **G93 F:** Feed rate in **1/min** (inverse time feed rate)
- **G94 F:** Constant speed and feed rate in **mm/min**
- **G95 F:** Constant speed and feed rate in **mm/revolution**

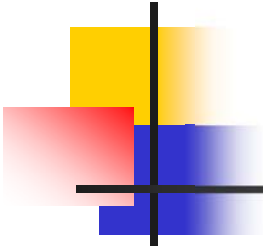
S: Speed

- **G96 S:** Constant cutting velocity in **m/min**
- **G97 S:** Constant cutting velocity in **rev/min**

Ex.: G97 S1000

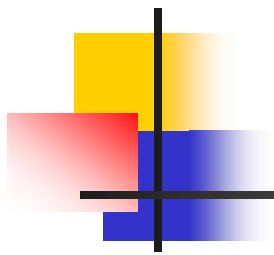


3- PREDEFINED SWITCHING AND AUXILIARY FUNCTIONS

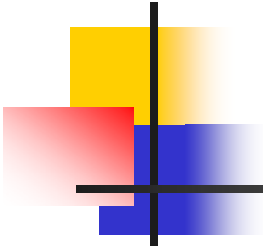


M: Miscellaneous Function

- **M3:** Sense of rotating main spindle **clock wise**.
- **M4:** Sense of rotating main spindle **counter clock wise**.
- **M5:** Spindle **stop** after motion been completed
- **M6:** Tool change
- **M8:** Coolant on
- **M9:** Coolant off
- **M30:** End of program

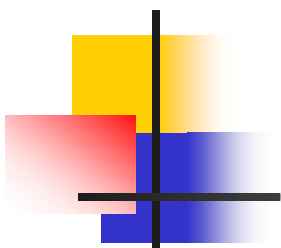


4- TOOL OFFSETS AND COMPENSATIONS



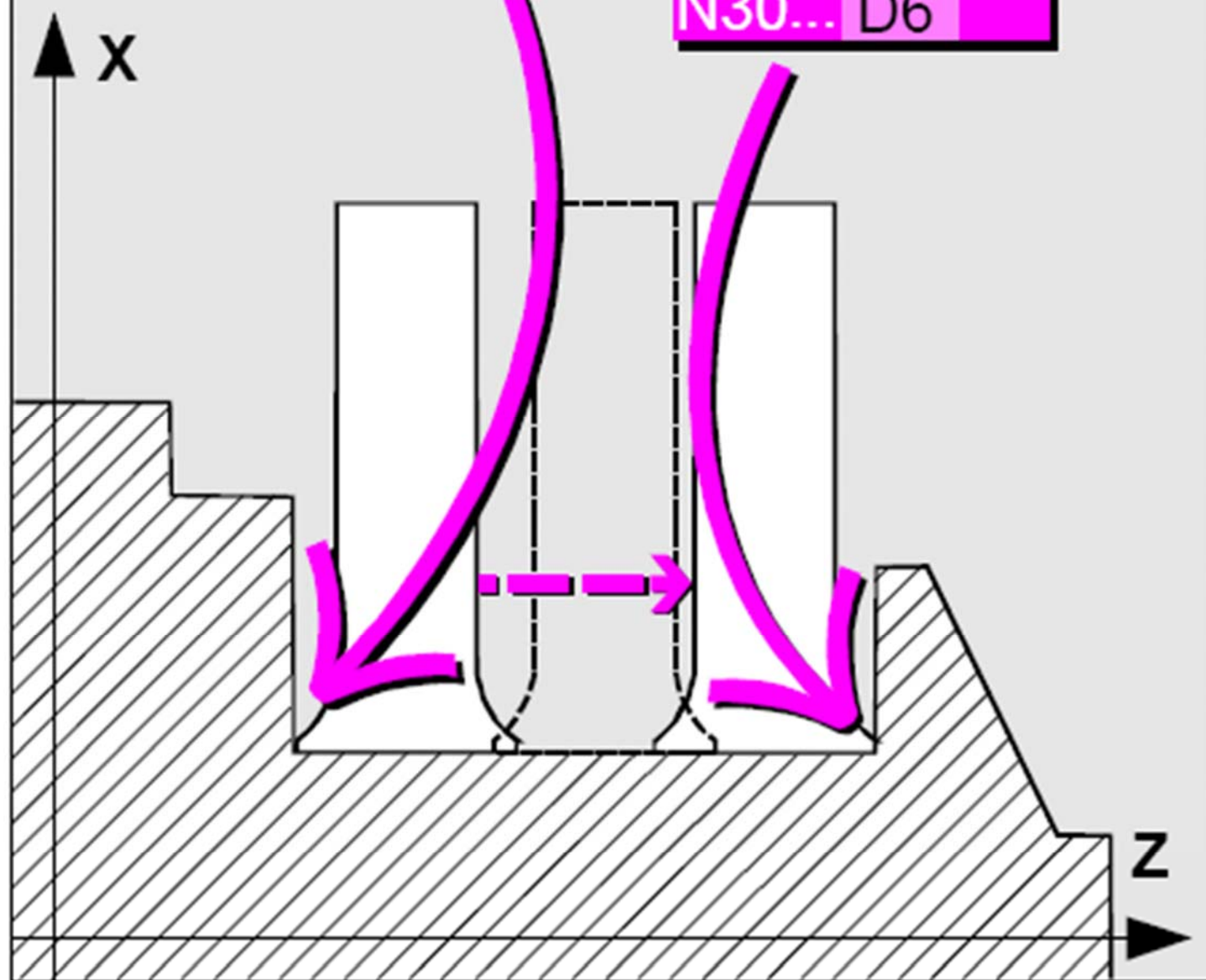
Tool call

- **T...** Call tool number
- **D...** Call tool offset, activate tool length compensation
- The tool length compensation acts automatically after tool offset **D** has been called.

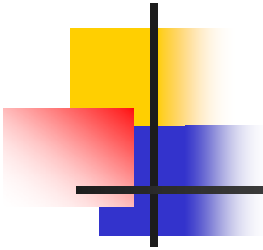


N10 G18 T17 D8

N30... D6

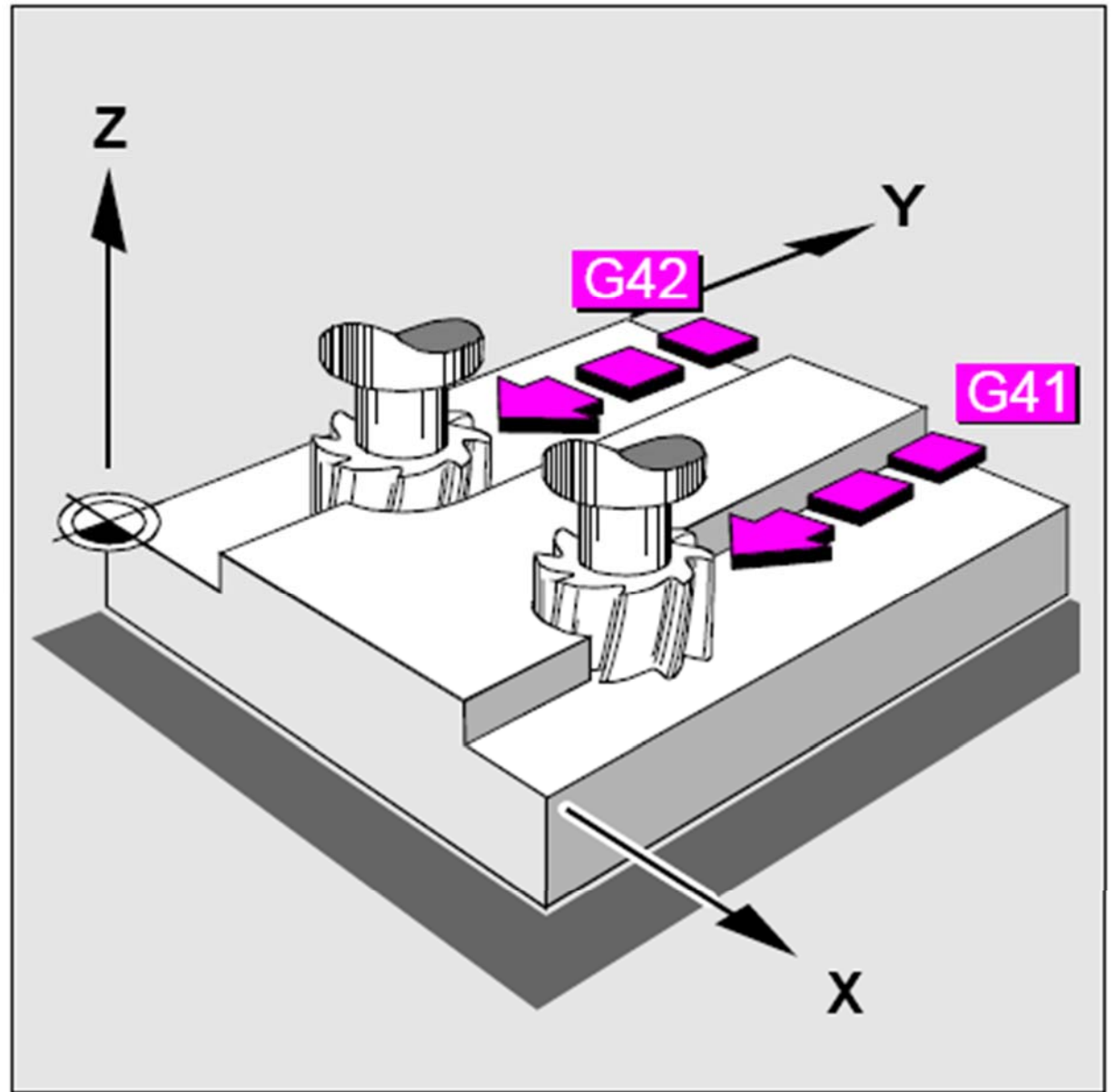


Offset values for left-hand and right-hand tool nose for recessing tool

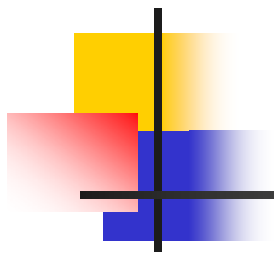


Tool/Cutter radius path compensation, G41/G42/G40

- **G41:** Call for cutter path compensation, tool in traversing direction **left of contour**
- **G42:** Call for cutter path compensation, tool in traversing direction **right of contour**
- **G40:** **Deselection** of cutter path compensation
- The offset acts only in the programmed working plane (G17 to G19).

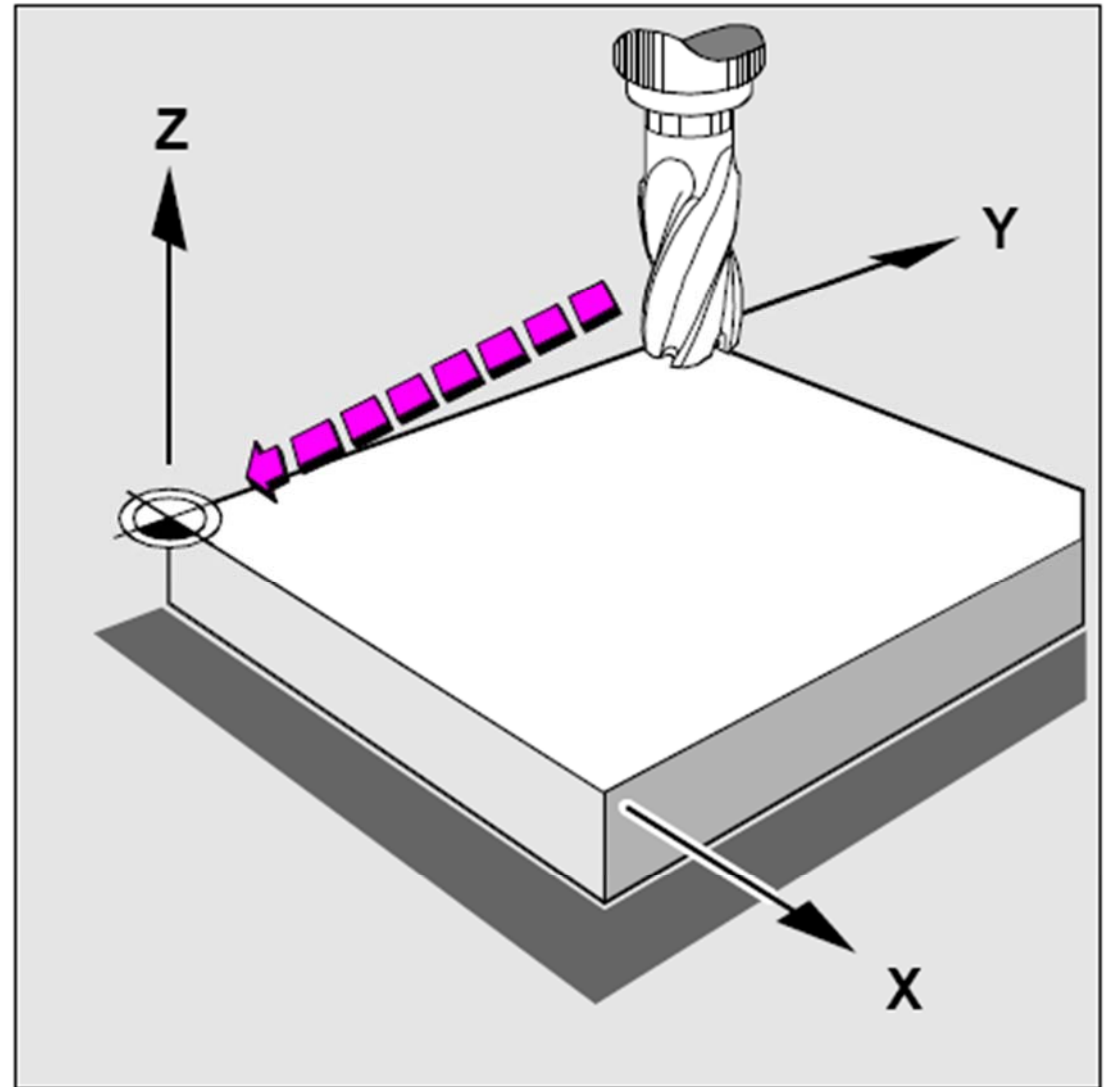


The control computes the tool path



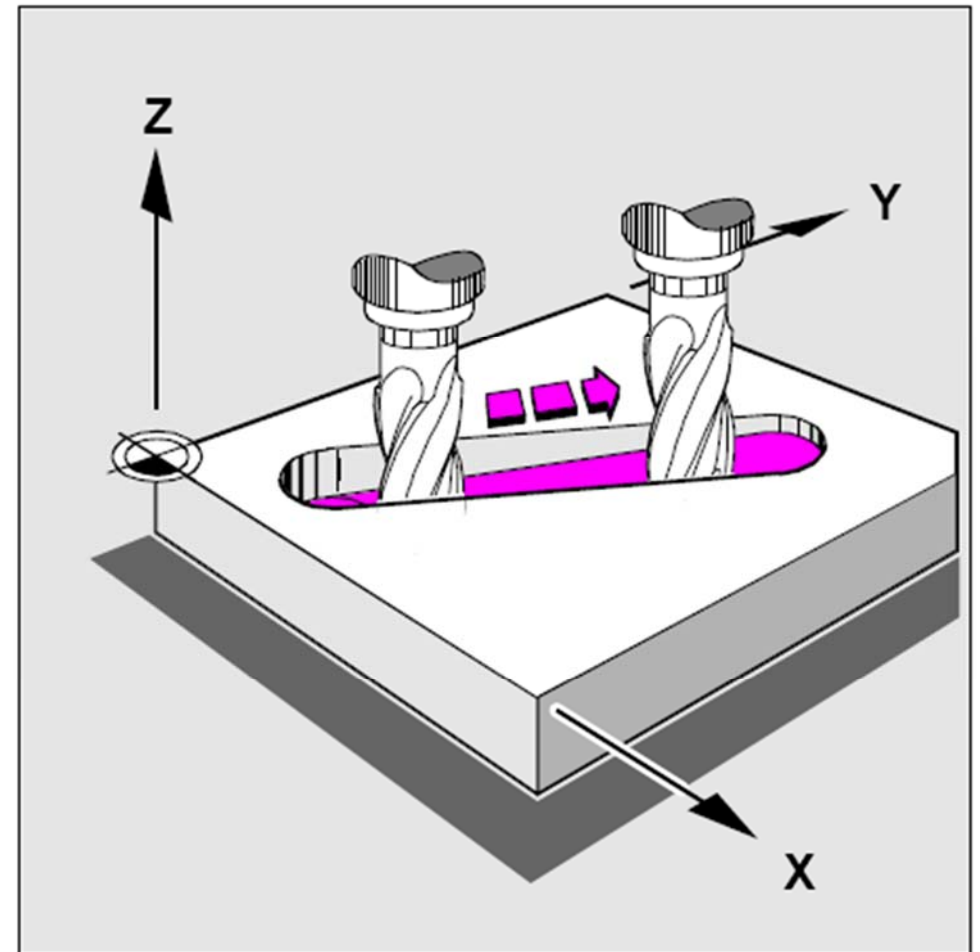
5- PROGRAMMING MOTION COMMANDS

Rapid traverse, G0

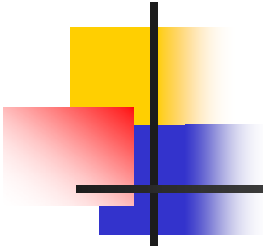


Fast tool positioning in rapid traverse for milling

Linear interpolation, G1



Making an inclined slot



Circular interpolation, **G2/G3**

- With **G2** the tool travels clockwise
- With **G3** counterclockwise.

There are two methods of input data:

1- Using Radius of the circular shape

2- Using Coordinates of the circular shape

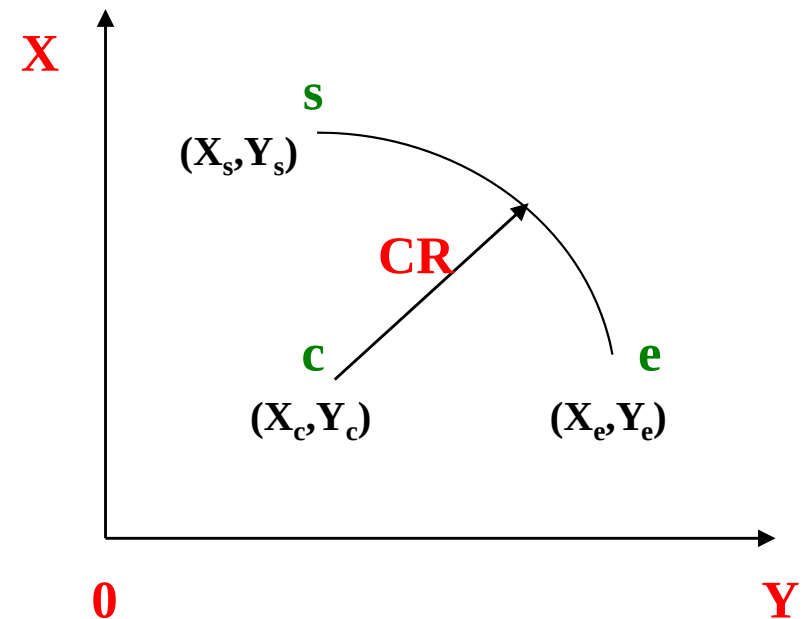
Circular interpolation, G2/G3

1st - Using Radius of the circular shape

The shape of this function:

G2 X 000 Y 000 CR000 F000

Ex. G2 X10 Y30 CR50 F100



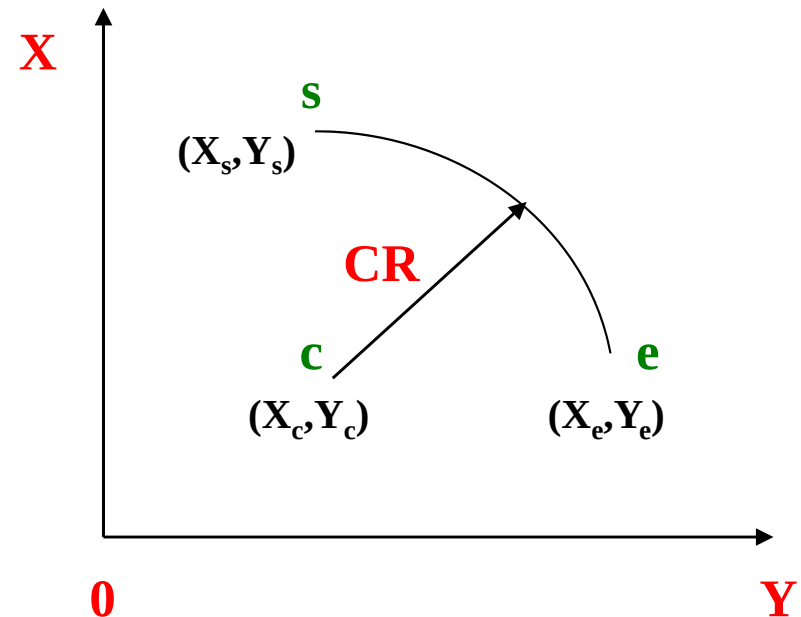
Circular interpolation, G2/G3

2- Using Coordinates of the circular shape

Where:

I: X_{center}

J: Y_{center}

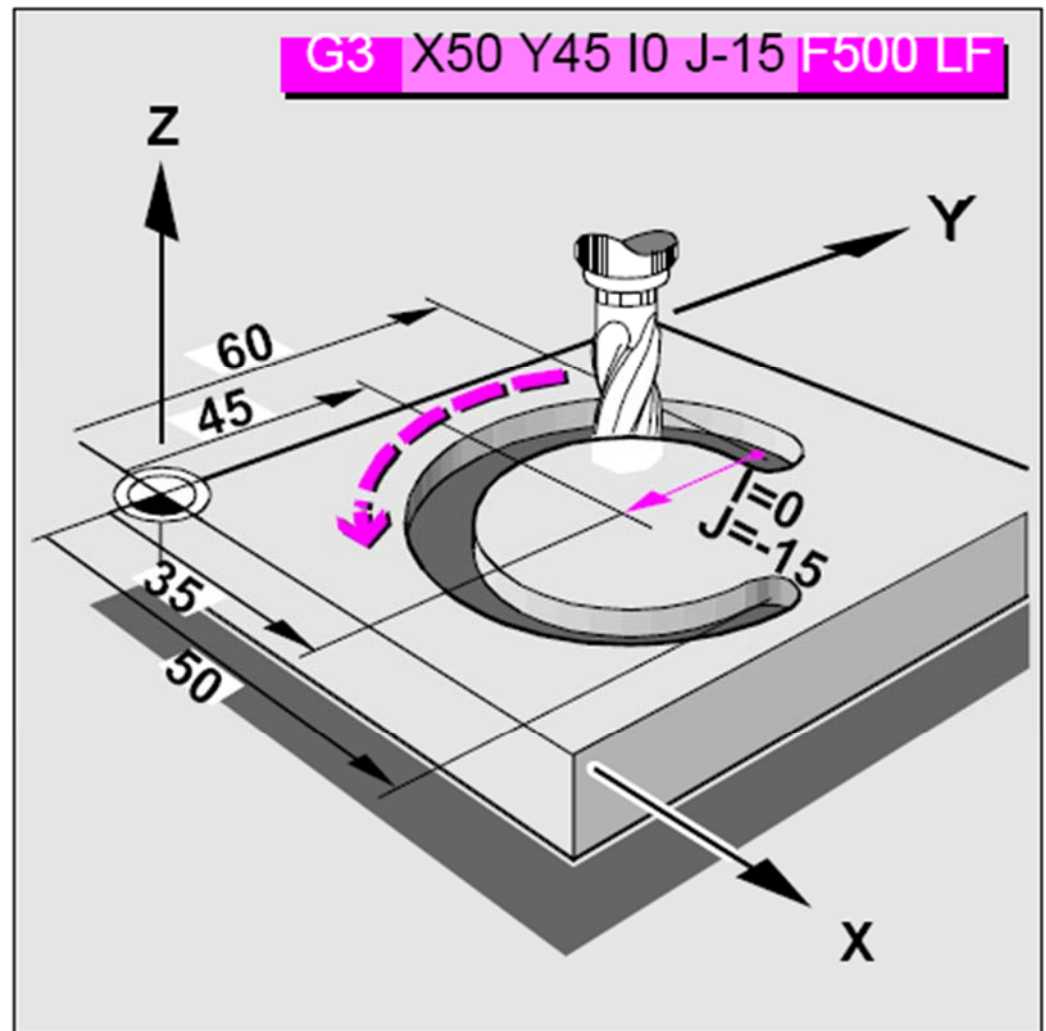


The shape of this function:

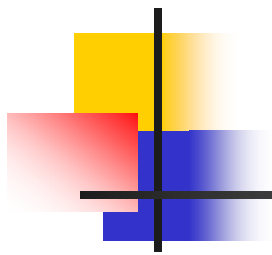
G3 X 000 Y 000 I000 J000 F000

Ex. G2 X50 Y45 I0 J-15 F500

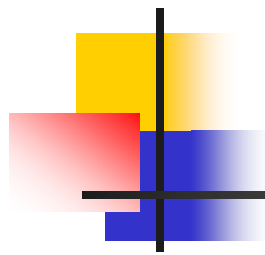
Circular interpolation, G2/G3



Making a circular slot

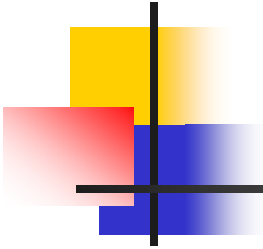


MOST USED G – CODES IN PROGRAMMING



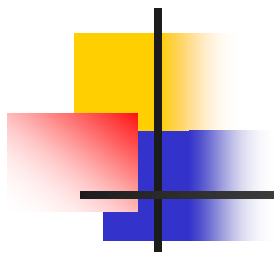
G - Codes

G0	Rapid traverse
G1	Machining traverse
G2	Circular interpolation clockwise
G3	Circular interpolation counterclockwise
G4	Dwell time
G17	Working plane selection XY
G18	Working plane selection XZ
G19	Working plane selection YZ
G40	Cutter radius compensation OFF
G41	Cutter radius compensation ON Left
G42	Cutter radius compensation On Right

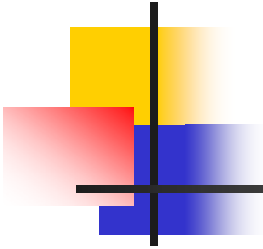


G – Codes, cont.

G53	Deselecting settable zero offset
G54-G57	Settable zero offset
G70	Input system inch
G71	Input system metric
G90	Absolute dimensioning
G91	Incremental dimensioning
G94	Feed in mm/min, inch/min
G95	Rotational feed rate in mm/rev, inch/rev
G96	speed (m/min)
G97	speed (rev/min)

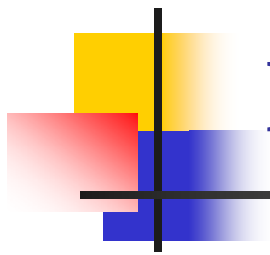


MOST USED M – CODES IN PROGRAMMING



M - Codes

M0	Programmed stop
M1	Optional stop
M2	End of program
M3	Spindle on clockwise
M4	Spindle on counterclockwise
M5	Spindle stop



M - Codes

M6	Tool change
M8	Coolant on
M9	Coolant off
M17	Subprogram end
M30	Program end



END